

Mathematical Modelling of Volume Expansion in Silicon–Graphite Composite Anodes for Li-ion Batteries

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1 Introduction

The anode in modern lithium-ion batteries is typically made from graphite, which intercalates lithium ions between its graphene layers during charging. While graphite is electrochemically stable, its theoretical specific capacity is limited to 372 mA h g^{-1} . Silicon has a much higher theoretical capacity of 3579 mA h g^{-1} because it alloys with up to 3.75 lithium atoms per silicon atom. However, this alloying reaction comes with a large volume expansion that can reach around 275%, compared to roughly 10% for graphite.

This volume change is important because it leads to mechanical degradation of the electrode, cracking of the solid electrolyte interface (SEI), and eventually capacity fading. A common approach in industry is to blend a small amount of silicon into the graphite anode to get a capacity boost without too much mechanical stress. In this work, we develop a mathematical model that predicts the volume ratio V/V_0 of a SiGr composite anode as a function of state of charge (SOC) for different silicon weight fractions. The model uses only crystallographic parameters and does not rely on any fitted experimental data.

2 Theoretical Background

2.1 Graphite Intercalation

Graphite has a hexagonal layered crystal structure. In its pristine state, the graphene layers are held together by van der Waals forces with an interlayer spacing of $d_0 = 3.354 \text{ \AA}$ [1]. During lithiation, Li^+ ions insert between the graphene layers. At full lithiation the compound LiC_6 is formed, where every interlayer gallery contains lithium, and the spacing increases to $d_{\text{Li}} = 3.706 \text{ \AA}$ [2].

The in-plane lattice parameters (a , b axes) do not change significantly during lithiation, so the volume change is dominated by the c -axis expansion. The maximum fractional volume change is therefore

$$\alpha_{\text{Gr}} = \frac{d_{\text{Li}} - d_0}{d_0} = \frac{3.706 - 3.354}{3.354} = 0.1049 \approx 10.5\% \quad (1)$$

Lithiation does not happen uniformly. Instead, graphite goes through a series of *staging transitions*, where lithium fills every n -th gallery [1]. The main stages are:

- **Stage 4** ($n = 4$): every 4th gallery occupied, at low SOC
- **Stage 3** ($n = 3$): every 3rd gallery
- **Stage 2** ($n = 2$): every 2nd gallery, at intermediate SOC
- **Stage 1** ($n = 1$): every gallery occupied, corresponding to LiC_6

At stage n , the average interlayer spacing is

$$\bar{d}(n) = d_0 + \frac{1}{n} (d_{\text{Li}} - d_0) \quad (2)$$

and the corresponding fractional volume change is

$$\left. \frac{\Delta V}{V_0} \right|_{\text{stage } n} = \frac{d_{\text{Li}} - d_0}{n \cdot d_0} \quad (3)$$

Using equation (3), the volume levels at each stage are computed in Table 1.

Table 1: Graphite volume expansion at each lithiation stage, calculated from the interlayer d-spacing formula (Eq. 3).

Stage n	$\bar{d}$ [Å]	$\Delta V/V_0$	Incremental step
4	3.442	2.62%	—
3	3.471	3.50%	0.88%
2	3.530	5.25%	1.75%
1	3.706	10.49%	5.24%

The largest single jump occurs at the stage 2 to stage 1 transition, which accounts for about half of the total expansion. This is consistent with the observation that graphite anodes swell most at high SOC [6].

2.2 Silicon Alloying

Silicon stores lithium through an alloying reaction rather than intercalation. At maximum lithiation, the compound $\text{Li}_{15}\text{Si}_4$ (equivalently $\text{Li}_{3.75}\text{Si}$) is formed [3]. The volume expansion is calculated from the crystal structures of the two end-member phases.

Pure silicon has a diamond cubic structure with lattice parameter $a_{\text{Si}} = 5.431 \text{ Å}$ and 8 atoms per unit cell, giving a volume per Si atom of

$$V_{\text{Si}} = \frac{a_{\text{Si}}^3}{8} = \frac{(5.431)^3}{8} = 20.02 \text{ Å}^3 \quad (4)$$

The fully lithiated phase $\text{Li}_{15}\text{Si}_4$ has a BCC structure with $a_{\text{Li}_{15}\text{Si}_4} = 10.63 \text{ Å}$ and 4 formula units per cell, which means 16 Si atoms per cell [4]. The volume per Si atom in the lithiated phase is

$$V_{\text{Li}_{15}\text{Si}_4} = \frac{a_{\text{Li}_{15}\text{Si}_4}^3}{16} = \frac{(10.63)^3}{16} = 75.07 \text{ Å}^3 \quad (5)$$

The maximum fractional volume change is then

$$\alpha_{\text{Si}} = \frac{V_{\text{Li}_{15}\text{Si}_4} - V_{\text{Si}}}{V_{\text{Si}}} = \frac{75.07 - 20.02}{20.02} = 2.752 \approx 275\% \quad (6)$$

This means silicon nearly quadruples in volume at full lithiation, which is about 26 times larger than the graphite expansion. The lithiation of silicon proceeds in two main stages [5]:

1. Formation of amorphous $a\text{-Li}_x\text{Si}$ at low to moderate lithiation, contributing roughly 40% of the total expansion.
2. Nucleation of crystalline $c\text{-Li}_{15}\text{Si}_4$ at high lithiation, contributing the remaining 60%.

2.3 Composite Anode: Rule of Mixtures

We consider a composite anode consisting of silicon with weight fraction w_{Si} and graphite with weight fraction $(1 - w_{\text{Si}})$. To compute the volume change we first convert to volume fractions using the material densities ($\rho_{\text{Si}} = 2.329 \text{ g cm}^{-3}$, $\rho_{\text{Gr}} = 2.260 \text{ g cm}^{-3}$):

$$\phi_{\text{Si}} = \frac{w_{\text{Si}}/\rho_{\text{Si}}}{w_{\text{Si}}/\rho_{\text{Si}} + (1 - w_{\text{Si}})/\rho_{\text{Gr}}} \quad (7)$$

The composite volume ratio as a function of SOC is then given by the Voigt rule of mixtures, assuming each phase expands independently:

$$\boxed{\frac{V}{V_0}(z) = 1 + \phi_{\text{Si}} \frac{\Delta V_{\text{Si}}}{V_0}(z) + (1 - \phi_{\text{Si}}) \frac{\Delta V_{\text{Gr}}}{V_0}(z)} \quad (8)$$

where $z \in [0, 1]$ is the anode SOC and $\Delta V_i/V_0(z)$ is the fractional volume change of each pure material at that SOC.

3 Mathematical Model

3.1 Sigmoid Approximation for Phase Transitions

The staging transitions in graphite and the two-phase lithiation of silicon are discrete jumps in practice. To model these mathematically, we use sigmoid functions as smooth step approximations [7]:

$$\sigma(z, z_0, s) = \frac{1}{1 + \exp\left(-\frac{z - z_0}{s}\right)} \quad (9)$$

where z_0 is the SOC at the centre of the transition and s controls the width (related to thermal broadening of the first-order phase transition). The expansion of a material is then written as a weighted sum of sigmoids, one per transition.

3.2 Graphite Expansion Model

From Table 1, we group the staging transitions into three and compute the sigmoid weights directly from the incremental volume steps:

$$\frac{\Delta V_{\text{Gr}}}{V_0}(z) = \alpha_{\text{Gr}} \cdot \frac{\sum_{i=1}^3 w_i \sigma(z, z_i, s_i) - f(0)}{\sum_{i=1}^3 w_i \sigma(1, z_i, s_i) - f(0)} \quad (10)$$

where $f(0) = \sum_i w_i \sigma(0, z_i, s_i)$ ensures $\Delta V(0) = 0$, and the normalisation ensures $\Delta V(1) = \alpha_{\text{Gr}}$ exactly. The parameters are:

Table 2: Sigmoid parameters for graphite staging model. Weights w_i are derived from the d-spacing formula.

i	Transition	z_i	s_i	w_i
1	Empty $\rightarrow$ Stage 3	0.12	0.04	0.334
2	Stage 3 $\rightarrow$ Stage 2	0.50	0.06	0.167
3	Stage 2 $\rightarrow$ Stage 1	0.90	0.04	0.499

The weights are computed as $w_i = \Delta_i / \alpha_{\text{Gr}}$, where Δ_i is the incremental volume step for each transition from Table 1. For example, the first weight is $w_1 = 0.0350 / 0.1049 = 0.334$, corresponding to the volume jump when reaching stage 3.

3.3 Silicon Expansion Model

Silicon lithiation is modelled with two sigmoid steps representing the amorphous and crystalline phases:

$$\frac{\Delta V_{\text{Si}}}{V_0}(z) = \alpha_{\text{Si}} \cdot \frac{\sum_{j=1}^2 w_j \sigma(z, z_j, s_j) - g(0)}{\sum_{j=1}^2 w_j \sigma(1, z_j, s_j) - g(0)} \quad (11)$$

Table 3: Sigmoid parameters for silicon expansion model.

j	Phase	z_j	s_j	w_j
1	Amorphous $a\text{-Li}_x\text{Si}$	0.30	0.10	0.40
2	Crystalline $c\text{-Li}_{15}\text{Si}_4$	0.75	0.08	0.60

The broader sigmoid widths compared to graphite ($s = 0.08\text{--}0.10$ vs. $0.04\text{--}0.06$) reflect the fact that amorphous-phase transitions do not have sharp crystallographic boundaries.

4 Results

Using the weight fractions $w_{\text{Si}} \in \{0, 0.15, 0.30, 0.45, 0.60, 0.75\}$, we compute V/V_0 over the full SOC range using Eq. (8). Table 4 shows the maximum expansion at full lithiation.

Table 4: Predicted anode volume ratio at full lithiation ($z = 1$).

w_{Si} [%]	V/V_0 at $z = 1$	Expansion [%]
0	1.105	10.5
15	1.492	49.2
30	1.882	88.2
45	2.275	127.5
60	2.672	167.2
75	3.073	207.3

The relationship between silicon content and maximum expansion is nearly linear because the densities of silicon and graphite are similar ($\rho_{\text{Si}} \approx \rho_{\text{Gr}}$), so the weight fraction is close to the volume fraction. The expansion at 15 wt% silicon is already about 5 times larger than pure graphite, which shows why even small amounts of silicon have a significant mechanical impact on the electrode.

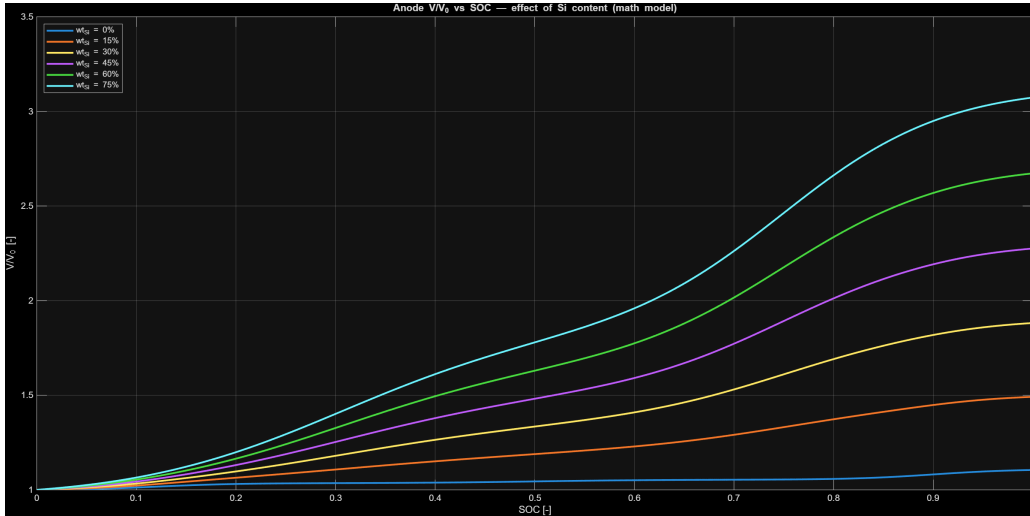


Figure 1: Anode volume ratio V/V_0 vs. SOC for different silicon weight fractions, computed from the mathematical model.

5 Summary

We presented a mathematical model for predicting the volume expansion of silicon–graphite composite anodes as a function of SOC. The model is based on two crystallographic facts:

1. The interlayer spacing of graphite increases from 3.354 Å to 3.706 Å upon full lithiation, giving a 10.5% expansion.

2. The atomic volume of silicon increases from $20.02 \text{ \AA}^3$ to $75.07 \text{ \AA}^3$ upon forming $\text{Li}_{15}\text{Si}_4$, giving a 275% expansion.

Phase transitions are approximated with sigmoid functions whose weights are derived from the staging d-spacing formula (not from curve fitting). The composite expansion follows from a simple rule of mixtures. The model requires no experimental data and can be used as a baseline for evaluating electrode designs with different silicon contents.

References

- [1] J. R. Dahn, “Phase diagram of Li_xC_6 ,” *Physical Review B*, vol. 44, no. 17, pp. 9170–9177, 1991.
- [2] T. Ohzuku, Y. Iwakoshi, and K. Sawai, “Formation of lithium-graphite intercalation compounds in nonaqueous electrolytes and their application as a negative electrode for a lithium ion (shuttlecock) cell,” *Journal of The Electrochemical Society*, vol. 140, no. 9, pp. 2490–2498, 1993.
- [3] M. N. Obrovac and L. Christensen, “Structural changes in silicon anodes during lithium insertion/extraction,” *Electrochemical and Solid-State Letters*, vol. 7, no. 5, pp. A93–A96, 2004.
- [4] M. Zeilinger, D. Benson, U. Häussermann, and T. F. Fässler, “Single crystal growth and thermodynamic stability of $\text{Li}_{15}\text{Si}_4$,” *Chemistry of Materials*, vol. 25, no. 9, pp. 1960–1967, 2013.
- [5] M. N. Obrovac and V. L. Chevrier, “Alloy negative electrodes for Li-ion batteries,” *Chemical Reviews*, vol. 114, no. 23, pp. 11444–11502, 2014.
- [6] Y. Qi and S. J. Harris, “In situ observation of strains during lithiation of a graphite electrode,” *Journal of The Electrochemical Society*, vol. 157, no. 6, pp. A741–A747, 2010.
- [7] G. L. Plett, *Battery Management Systems, Volume I: Battery Modeling*. Artech House, 2015.